东 南 大 学 考 试 试 卷（B 卷）

课程代码	BJSL0041	课程名称	离散数学	考试学期	2022-2023-3
适用专业	计算机类	考试形式	闭 卷	考试时长	120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

自觉遵守考场纪律 如考试作弊 此答卷无效

姓名_____ 学号_____

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题目	1	2	3	4	5	6	7	8	9	10	总分
得分											
批阅人											

1. Choice questions (2 points for each question)

(1). Which of the following is a tautology? ( )

A. $P \to P \wedge Q$.

B. $P \vee Q \to P$.

C. $P \to (\neg P \to Q)$.

D. $(\neg P \to Q) \to P$.

(2). Which of the following is NOT logically equivalent? ( )

A. $\exists x\, P(x) \wedge \exists x\, Q(x)$ and $\exists x\, (P(x) \wedge Q(x))$.

B. $\forall x\, p(x) \wedge A$ and $\forall x\, (p(x) \wedge A)$, where A is a proposition not involving quantifiers.

C. $\exists x\, p(x) \wedge A$ and $\exists x\, (p(x) \wedge A)$, where A is a proposition not involving quantifiers.

D. $\forall x\, P(x) \wedge \forall x\, Q(x)$ and $\forall x\, (P(x) \wedge Q(x))$.

(3). Which of the following set statements is wrong? ( )

A. $|P(\{\varnothing\})| = 2$.

B. $\varnothing \in \varnothing$.

C. $|A| - |B| \leq |A - B|$.

D. $A \times (B - C) = (A \times B) - (A \times C)$.

(4). Determine which of these functions from Z to Z is not injective. ( )

A. $f(x) = x-1$ B. $f(x) = x^2+1$

C. $f(x) = x^3$ D. $f(x) = 2x$

(5). For a Boolean algebra B containing n elements, which of the following can be a possible choice of n? ( )

A. 3 B. 4 C. 5 D. 6

(6). Let G be a group with identity element e, H is a subgroup of G, and $Ha = Hb$ for

some $a, b \in G$. Which of the following statement is true? ( )

A. $a = b$ B. $e \in Ha$

C. $ha = hb$ for some $h \in H$ D. $a \in Hb$

(7). If G is a non connected undirected graph with 36 edges, what is the minimum number of vertices in the graph? ( )

A. 7 B. 8 C.9 D. 10

(8). Which of the following conclusions is correct about the adjacency matrix of a graph. ( )

A. The adjacency matrix of a digraph is always asymmetric.

B. The adjacency matrix of a digraph can be symmetric or asymmetric.

C. The adjacency matrix of undirected graphs is always asymmetric.

D. The adjacency matrix of an undirected graph can be asymmetric or symmetric.

2. Completion (2 points for each question)

(1). Given the predicate formula ∀x P(x, y), the free variable is _____ (Fill in x, y, or null); given the predicate formula ∀x (P(x)∧Q(x)) ∨ ∀y R(y), the free variable is _____(Fill in x, y, or null). (1 point for each blank)

(2). In the domain of all books, let H(x) be "x is heavy" and C(x) be "x is confusing", then in ordinary English, (∀x) (H(x)→C(x)) means_____ _____ and (∃x) (C(x) ∧ H(x)) means _____. (1 point for each blank)

(3). Let A ={a, b}, how many distinct functions are there from A to A _____ , and how many distinct bijective functions are there from A to A _____ . (1 point for each blank)

(4). Given a Boolean algebra $< \mathcal{P}(S), \cap, \cup, \sim, \emptyset, S >$ with $S = \{a, b, c\}$, the complement element of $\{b\}$ is_____.

(5). Let $(N_8, +_8)$ be the group under the operation of module 8 addition. Is it a cyclic group?_____(Fill in "Yes" or "No") If yes, how many generators does it have?_____If no, explain the reason._____ (1 point for each blank)

(6). Complete bipartite graph $K_{m,n}$ has an Hamilton circuit, when m, n satisfy_____ _____

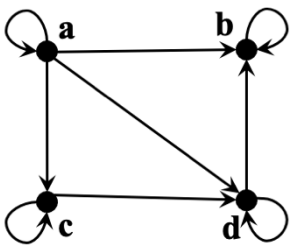
3. Prove that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$. (8 points)

4. We know that:
   (1) If Rama graduates, he will find a job;
   (2) If Rama finds a job, he will marry soon;
   (3) If Rama goes for higher degree, he will not get married;
   (4) Rama graduates and goes for higher degree.

   Prove all the four premises are not consistent. (10 points)

5. Let R_1 be a relations on A, relation $R=\{(a,b) \mid (b,a) \in R_1\}$, prove that R is an equivalence relation if R_1 is an equivalence relation. (9 points)

6. Let A={a, b, c, d}, R is a relation represented by the following directed graph.(9 points)



   (1) Find the transitive closure t(R) of R.
   (2) Determine whether the transitive closure t(R) is a partial order. If it is a partial order, draw the Hass diagram, find all the upper bounds and lower bounds of {c, d}.

7. Let $(Z,+)$ be the group of integers under the operation of addition. Determine whether the following sets can form a subgroup of $(Z,+)$. (9 points)
   (1) $S_1 = \{2n | n \in Z\}$.
   (2) $S_2 = \{2n+1 | n \in Z\}$.
   (3) $S_3 = \{-1, 0, 1\}$.

8. Let $N_4 = \{0,1,2,3\}$, $A = \{0,2,4,6\}$, $+_4$ and $+_8$ are the addition module 4 and 8, respectively. Let $f: N_4 \to A$ be a mapping defined by $f(x) = 2x$. (9 points)
   (1) Is $(N_4, +_4)$ a group? Justify your answer.
   (2) Is $(A, +_8)$ a group? Justify your answer.
   (3) Show that f is an isomorphism.

9. If G=(V, E) is a bipartite graph, and |V|=n, |E|=m. Show that $m \leq {n^2}/{2}$. (9 points)

10. Given a bipartite graph $G =< V_1, V_2, E >$, where V_1 and V_2 are vertex sets, and E is edge set. Show that if G is Hamilton graph, then $|V_1| = |V_2|$. (9 points)